

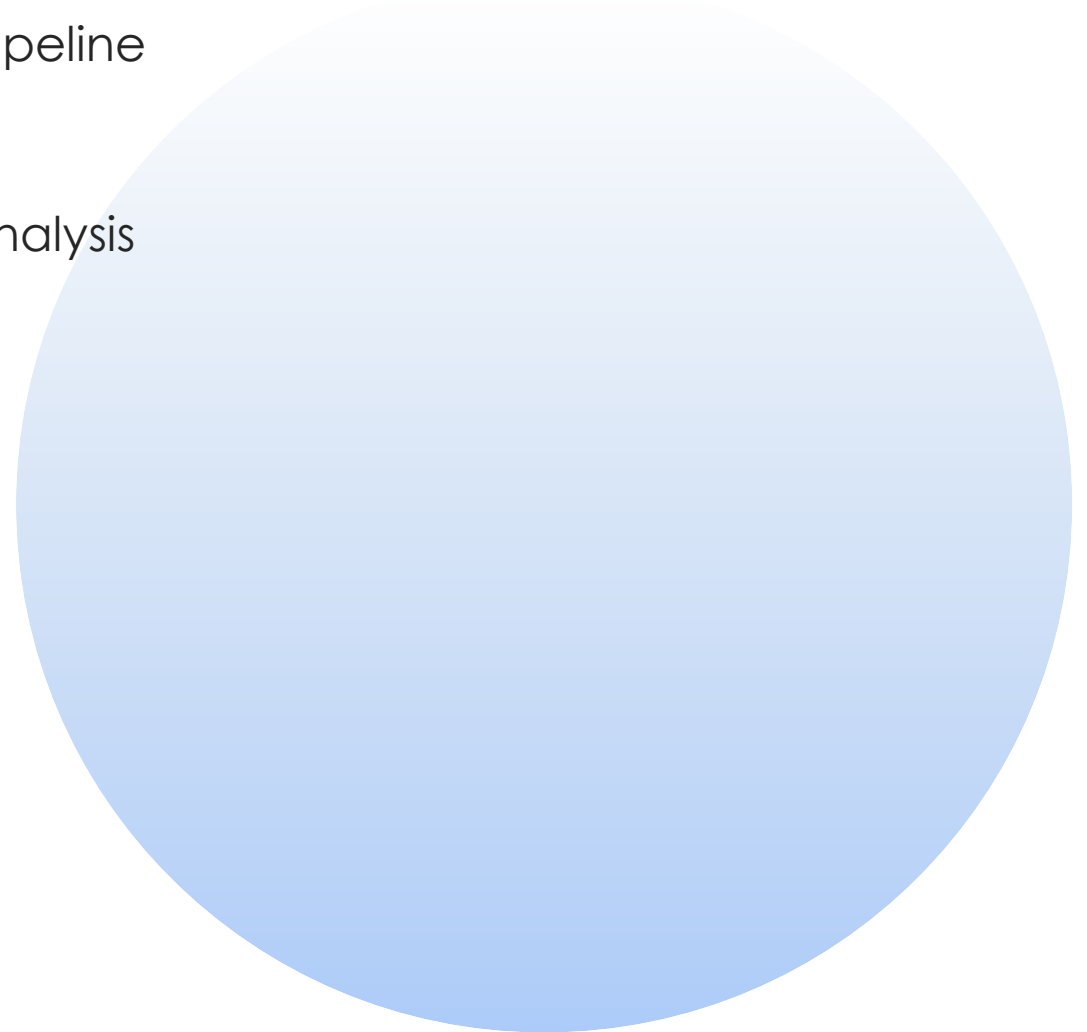
Practical Experiment: Compare the response speed of open-source Large Language Models (LLMs) to find the fastest model for real-time applications

Meeting 4 Create LLM Comparison Dashboards

Presented by: Mariyan Vasilev Apostolov
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Agenda

- Framework Architecture & Visualization Pipeline
- Bar Charts - Comparative Metric Analysis
- Line Charts - Performance Tracking
- Scatter Plots - Performance Distribution Analysis
- Pie Charts - Proportional Analysis
- Visualization Features
- Layout & Subplot Configuration
- Export & Reporting Capabilities
- Hierarchical Structure
- Model Evaluation Framework



Framework Architecture & Visualization Pipeline

Theoretical Foundation:

- Multi-dimensional performance assessment
- Cross-validation of model outputs
- Statistical significance visualization

Visualization Pipeline:

- **Data Ingestion:** CSV, BigQuery, or direct API
- **Preprocessing:** Aggregation, pivoting, normalization
- **Visual Mapping:** Metric → Chart Type Selection
- **Statistical Enhancement:** Error bars, confidence intervals
- **Export:** High-fidelity publication-ready figures

Methodological Rigor:

- Reproducible analysis pipeline
- Consistent metric scaling
- Quantitative performance assessment



Bar Charts - Comparative Metric Analysis

Applications:

- **Quantitative Comparison:** Side-by-side metric evaluation
- **Significance Testing:** Visual representation of variance
- **Ranking:** Clear performance ordering across models

Statistical Measures:

- **Mean Values:** Central tendency visualization
- **Standard Error:** Dispersion indication
- **Confidence Intervals:** Statistical significance

Metric Types:

- **BLEU Scores:** Translation quality assessment
- **Latency:** Computational efficiency
- **Token Counts:** Response complexity analysis

Interpretation Guidelines:

- Higher bars = Better performance
- Color coding = Model identification
- Error bars = Measurement uncertainty



Line Charts - Performance Tracking

Applications:

- **Performance Monitoring:** Continuous assessment
- **Trend Identification:** Improvement or degradation
- **Stability Analysis:** Variance over iterations

Analytical Components:

- **Trend Lines:** Direction of performance change
- **Markers:** Individual measurement points
- **Confidence Bands:** Uncertainty visualization

Statistical Metrics:

- **Slope Analysis:** Rate of performance change
- **Stationarity:** Performance consistency
- **Seasonality:** Cyclic performance patterns

Research Questions:

- Is model performance improving?
- How stable is the model over time?
- Are there performance fluctuations?

Scatter Plots - Performance Distribution Analysis

Statistical Analysis:

- **Distribution Patterns:** Data point clustering
- **Outlier Detection:** Anomaly identification
- **Density Estimation:** Performance concentration

Analytical Dimensions:

- **X-Axis:** Sequential test runs or latency
- **Y-Axis:** Performance metrics (BLEU, accuracy)
- **Color:** Model identification

Advanced Statistical Insights:

- **Bimodal Distributions:** Performance variability
- **Skewness:** Direction of performance bias
- **Kurtosis:** Outlier sensitivity

Research Applications:

- **Correlation Analysis:** Metric relationships
- **Regression:** Performance prediction
- **Outlier Investigation:** Anomalous results

Pie Charts - Proportional Analysis

Proportional Analysis:

- **Market Share:** Performance dominance visualization
- **Resource Allocation:** Efficiency assessment
- **Distribution:** Balanced vs. imbalanced performance

Statistical Metrics:

- **Proportional Representation:** Relative contributions
- **Percentage Distribution:** Normalized values
- **Central Tendency:** Dominant performance areas

Research Applications:

- **Portfolio Analysis:** Model performance distribution
- **Resource Optimization:** Allocation strategies
- **Performance Gaps:** Identifying improvement areas

Interpretation:

- Larger slices = Higher contribution
- Color coding = Model identification
- Percentage labels = Exact proportional values

Visualization Features

Statistical Enhancements:

- **Error Bars:** Measurement uncertainty
- **Confidence Intervals:** Statistical reliability
- **Trend Lines:** Analytical projections

Visual Enhancements:

- **Color Gradient:** Performance intensity
- **Alpha Blending:** Overlap visualization
- **Edge Enhancement:** Clarity improvement

Publication-Ready Features:

- **High DPI:** 300 dpi for academic journals
- **Tight Layout:** Optimized space utilization
- **Professional Colors:** Accessible design

Customization Options:

- **Metric Selection:** Specific performance indicators
- **Comparison Groups:** Model selections

Layout & Subplot Configuration

Organizing Multiple Charts

Grid System:

- Dynamic row/column calculation
- Max 3 columns per row
- Automatic subplot creation

Space Management:

- `figsize = (15, 5 * n_rows)`
- Tight layout optimization
- Consistent spacing between subplots

Rotation & Labels:

- 45-degree rotation for readability
- Right alignment for labels
- Title format: `{METRIC.upper()} {CHART_TYPE}`



Export & Reporting Capabilities

Professional Output Generation

Export Options:

- **PNG Format:** High-quality images
- **Custom DPI:** 72 (web) to 300 (print)
- **Tight Layout:** Optimized whitespace

File Structure:

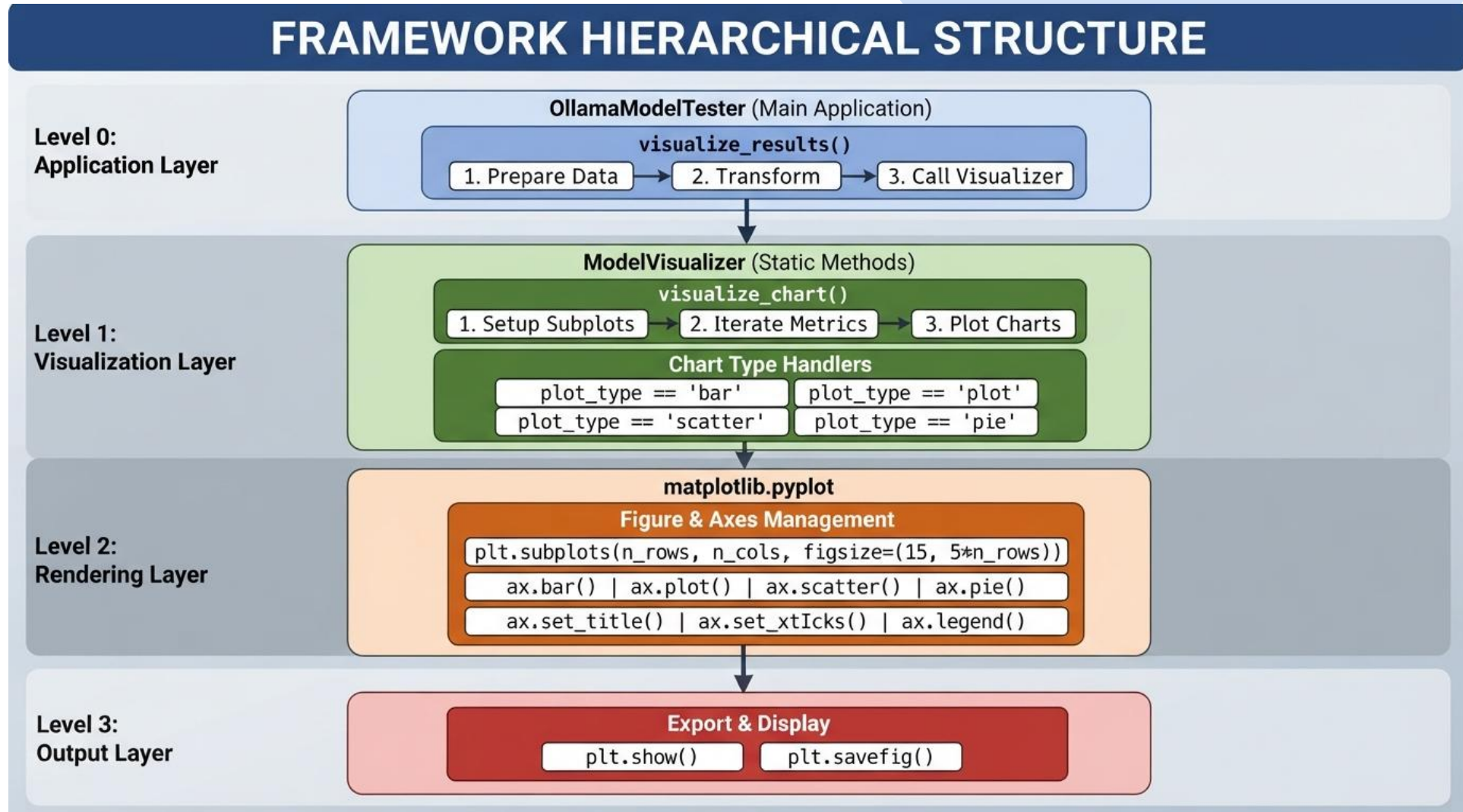
- Automatic folder creation
- Timestamp inclusion in filenames
- Organized chart types in subfolders

Use Cases:

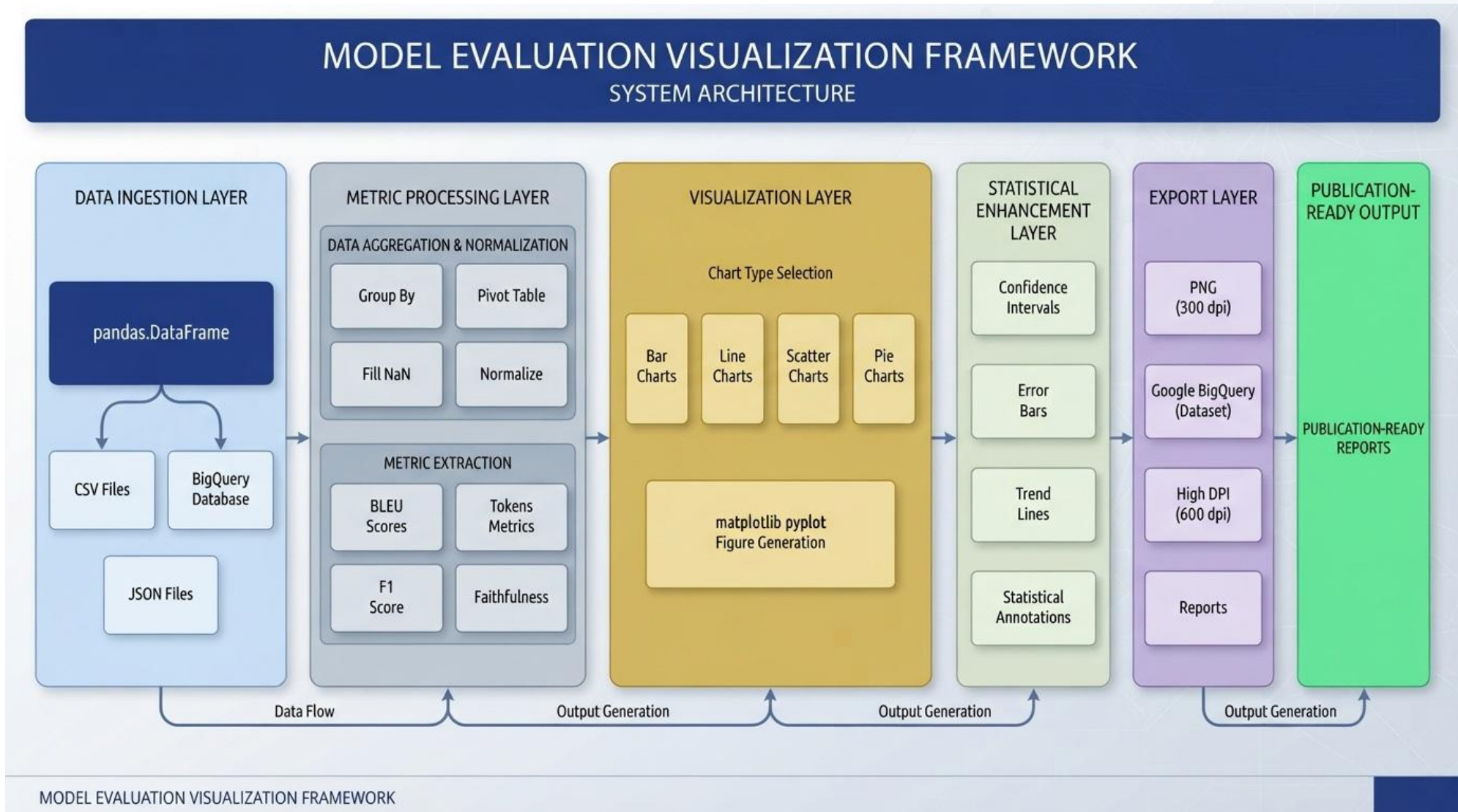
- **Reports:** Include in documentation
- **Presentations:** High-resolution slides
- **Dashboards:** Real-time monitoring



11 Hierarchical Structure

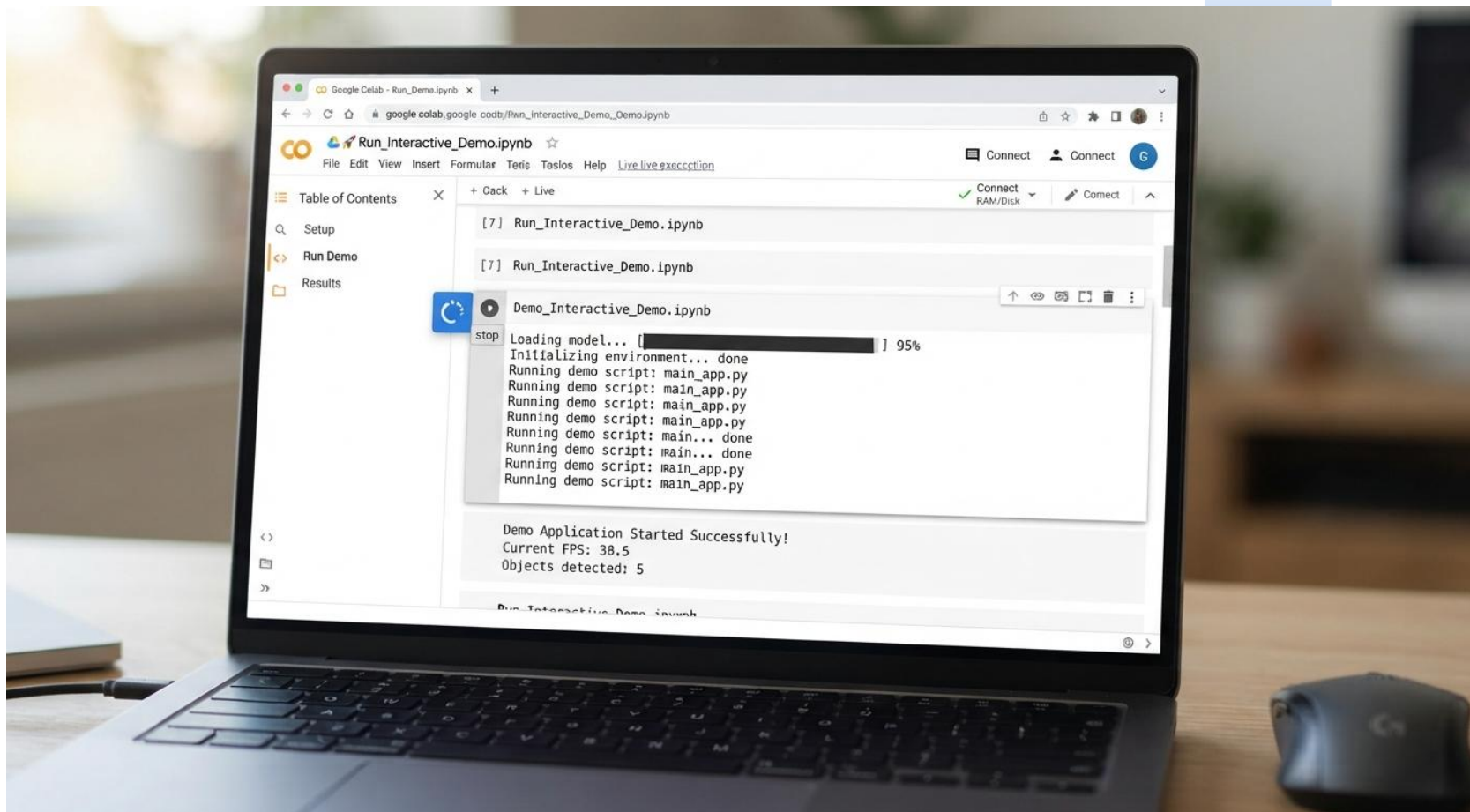


Model Evaluation Framework

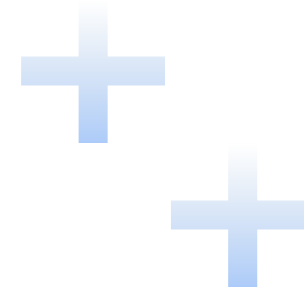


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LIVE DEMO

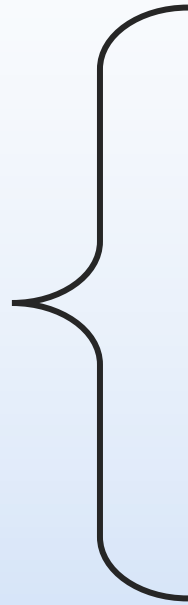


References



- Pandas DataFrame – <https://pandas.pydata.org/pandas-docs/stable/reference/api/pandas.DataFrame.html>
- Matplotlib Pyplot Bar – https://matplotlib.org/stable/api/_as_gen/matplotlib.pyplot.bar.html
- Matplotlib Pyplot Plot – https://matplotlib.org/stable/api/_as_gen/matplotlib.pyplot.plot.html
- Matplotlib Pyplot Scatter – https://matplotlib.org/stable/api/_as_gen/matplotlib.pyplot.scatter.html
- Matplotlib Pyplot Pie – https://matplotlib.org/stable/api/_as_gen/matplotlib.pyplot.pie.html

Thank you



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